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Exploiting multiple question factors for knowledge tracing

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ABSTRACT

Knowledge Tracing (KT) aims to predict future students' performance via their responses to a sequence of questions, which serves as a fundamental task for intelligent education. Most of the existing efforts directly predict students' performance depending on their dynamically changing knowledge states. However, the individualization of questions is neglected and difficulty level differ from question to question, which would give some valuable clues to KT. Towards this end, in this paper, we propose a novel Multiple Question Factors for Knowledge Tracing (MQFKT) method, which fully exploits various question factors to generate better prediction. On one hand, calibrated student-concept connection space is established to obtain fine-grained response representations on questions according to the information of responses on questions. On the other hand, individualized difficulty levels with particular concept for different questions are introduced for improving the prediction performance. Extensive experiments on three datasets have shown that the MQFKT approach achieves more precise prediction of student performance and better interpretation of the model.

1. Introduction

With the flouring of online education service systems, it has become paramount to speed up digital transformation of education system to keep flourishing the education system (Abdelrahman et al., 2022; Liu et al., 2021a). The world has entered into an online learning phase (i.e., a fashionable style of learning to offer tremendous online resources, such as courses and questions). It is more convenient for students and teachers to learn and teach at home on online education platforms anytime and anywhere, overturning the restrictions of physical classrooms. Online learning systems provide diverse intelligent educational services with massive amount of available student-system interactions, which can be further exploited for offering students appropriate guidance in the process of online learning. As a powerful tool to deduce the mastery level of students over concepts, Knowledge Tracing (KT) is capable of providing personalized learning paths for students, thus improving learning efficiency. For example, the computer-aided educational system can recommend appropriate questions for students with the help of the knowledge diagnosis results, while teachers and parents could also provide personalized guidance to each student. At its core is to estimate how likely a student will answer the next question correctly on the basis of his answers to a sequence of questions (Liu et al., 2021; Lu et al., 2022).

Up to now, many outstanding existing research rely on Deep Neural Networks (DNNs) to predict the performance on future questions from the implicit estimated hidden knowledge state. The recent popularity of DNNs adopted as a backbone for KT systems owes to their powerful capability for fine-grained feature extraction and non-linearity transformation, making it suitable for modeling the sophisticated learning process in scenarios with a large amount of learning interaction data. Therefore, more and more attention are paid on Deep Knowledge Tracing to trace student's knowledge state by means of DNNs to effectively construct the computer-aided educational system. Generally speaking, there are two key components in learnable KT models (Long et al., 2021; Sun et al., 2021; Zhang et al., 2017) — 1) *write process*, which updates the *knowledge state* according to the correctness of the student's answer, and 2) *read process*, which reconstructs historical interactions mainly from current knowledge state.

Despite the success of the previous methods, we argue that most of them largely rely on the student's knowledge state, which are vulnerable to fine representations and difficulty level of questions. Actually, recent KT models, such as (He et al., 2022; Song et al., 2022; Wang, Ma, Zhao, Li, & He, 2022a; Wang, Ma, Zhao, Yang, & Chang, 2022c; Liu et al., 2020; Zhang et al., 2021), highlight the importance of question representation (aka. embedding) as another key factor along with knowledge state in predicting student performance. Typically, PEBG (Liu et al.,

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2020) utilizes the plentiful implicit information among questions and concepts to generate the question representation for improving the performance of the existing KT models. Furthermore, Bi-CLKT (Song et al., 2022) obtains the question representation by devising the node-level and graph-level contrastive learning on constructed question influence subgraph to enhance prediction performance. Nevertheless, these methods fail to exploit the response records sufficiently. While intuitively useful to integrate question factors into the knowledge tracing process, it is non-trivial to do it well.

To motivate our work, we conduct data analysis on ASSIST09 (details of data statistics is summarized in section 5.1). We report student distribution within different scale of question intervals in Fig. 2 and data distribution in student-concept connection space (the construction process will be presented in section 4) in Fig. 3 respectively. Concretely, due to the limited interaction between students and questions, we analyze the variation of students' number with different number of questions with increment 100 for ASSIST09 dataset in Fig. 2(a). In addition, considering that most students are concentrated on the first interval, we further divide the first interval with increment 10 in Fig. 2 (b) to better display the analysis results for exploring two types of students' response distribution. Similarly, considering the relatively small value of the student-concept connection space in Fig. 3(a), we analyze it with increment 10 in Fig. 3(b). Accordingly, we have some interesting observations: (1) The majority of students only answered the first interval in Fig. 2(a) and responded with a certain number of incorrect and correct answers on each subinterval of the first interval in Fig. 2(a) (shown in Fig. 2(b)), suggesting there is a wealth of information of responses to be revealed; (2) In terms of the student-concept connection space, the number of questions answered by a student on a specific concept (i.e., the value of a specific entry in student-concept connection space) is relatively concentrated (shown in Fig. 3(b)), which hints the potential of amending these values to enhance the reliability of students' cognitive state.

What is more, we recap the motivating example in Fig. 1 to explain the importance of integrating difficulty level of questions for KT. Concretely, student *Bob* gives different responses (i.e., correct and incorrect) on two consecutive questions (i.e., q_3 and q_4) with the same concepts, owing to the fact that each question has individualized difficulty level. To this end, we argue that it is beneficial to maximally extracting and exploiting plentiful underlying information between student-question answer space and question-concept association space and considering difficulty level on different questions at the same time.

Unfortunately, there are some challenges along this line. On one hand, the number of students' responses on the questions on a specific concept differ dramatically, and it is not an easy work to update these various values to attain the reliable student-concept connection space. On the other hand, introducing difficulty levels for different questions and ensuring interpretability of simulated interactions between students and questions are very challenging when predicting student performance. In light of the aforementioned challenges, we propose a general Multiple Question Factors for Knowledge Tracing (MQFKT) method, which generates fine response representations of each question with the revised student-concept connection space and considers the difficulty of question via interpretable interactions between students and questions. To be specific, from the perspective of fine-grained response representation, the reliability of two built student-concept connection spaces is separately modified based on the number of students' incorrect and correct responses on the questions corresponding to a specific concept. Afterwards, we acquire low-dimensional representations of two calibrated student-concept connection spaces with dimensionality reduction, so as to obtain distinct (i.e., incorrect and correct) response representations on each question. Next, in terms of prediction, we design a new interaction function to combine the student's knowledge state in KT with the inherent factors of question and use MultiLayer Perceptron (MLP) to model the complex interactions between students and questions. Therefore, with the help of fine response representations and sophisticated interactions between students and questions, we can predict student's performance more accurate in an explainable manner in future. We conduct extensive experiments on three datasets to clearly demonstrate the effectiveness of the MQFKT in student performance prediction and its superior interpretability. The key contributions of our paper are summarized as follows:

(1) We highlight the critical importance of explicitly exploiting the question factors to generate better predictions.
(2) We propose MQFKT, a new knowledge tracing method based on DNNs, which explicitly encodes various question factors in the form of representative embeddings and complicated interactions between students and questions.
(3) Extensive experiments on three real-world datasets validate the effectiveness and superior interpretability of MQFKT. Further investigations demonstrate the information of responses on questions and inherent factors of questions are meaningful and essential for predicting student's performance.

2. Related works

In this section, we first introduce existing related works about knowledge tracing, followed by presenting the question representation in detail according to the development timeline.

2.1. Knowledge tracing

There are many efforts in literature of knowledge tracing. Generally, existing methods can be classified into traditional methods and deep learning-based methods. Specifically, most traditional methods generate prediction based on the factors related to learning. In particular, BKT (Corbett & Anderson, 1995) is a classical knowledge tracing method using Hidden Markov Model which hypothesizes students' knowledge states as a set of binary variables and update them on the basis of Bayesian rules. Moreover, another typical type is Factor Analysis methods. KTM (Vie & Kashima, 2019) based on Factorization Machine, uses many types of features like questions, concepts to predict student's performance. In addition, ABKT (Liu et al., 2022) introduces the students' ability factor into learning feedback attribution and further analyses the learning process from knowledge and ability perspectives simultaneously.

Nevertheless, a majority of deep learning-based methods rely on the knowledge state which is represented as vector. One representative group in deep learning-based methods is single-state methods. To the

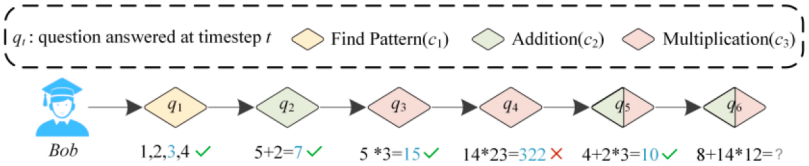
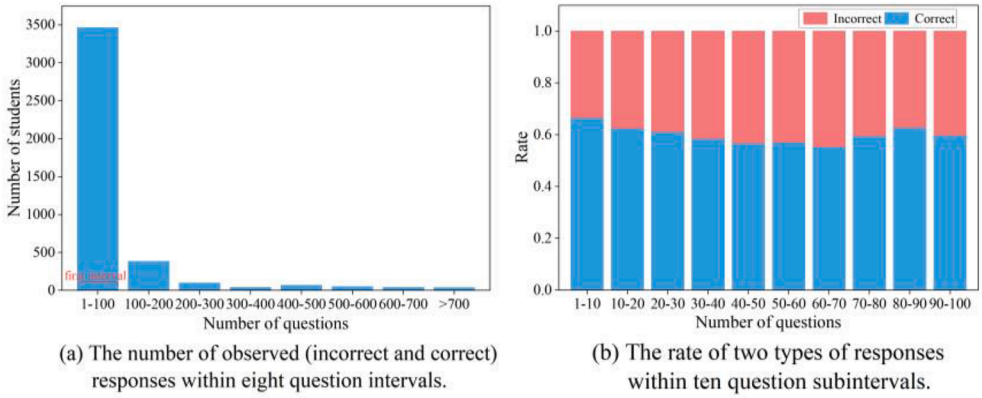


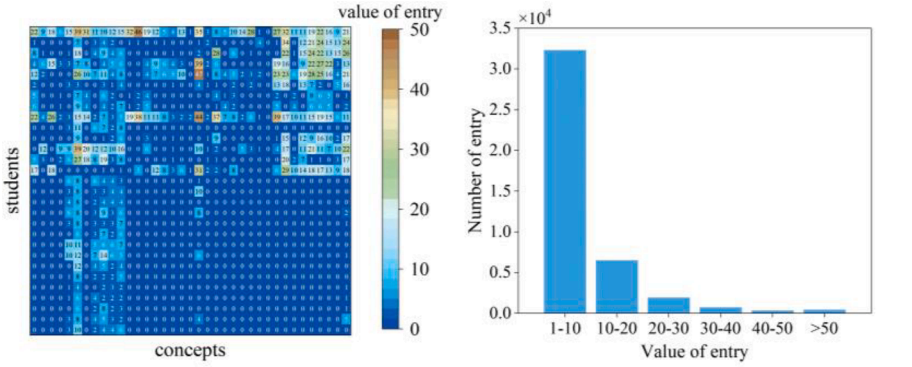
Fig. 1. (a) A student *Bob* first selects some questions (i.e., q_1, q_2, q_3, q_4 and q_5) to practice and leaves his response at different timesteps. The goal of KT is to use his historical question sequences to predict the probability that he correctly answers the next unobserved question (i.e., q_6). Different colored rhombuses represent different concepts.



(a) The number of observed (incorrect and correct) responses within eight question intervals.

(b) The rate of two types of responses within ten question subintervals.

Fig. 2. (a) The student distribution within different scale of question intervals on ASSIST09. The first interval is marked in red font. (b) The rate of two responses on each subinterval of the first interval in Fig. 2(a).



(a) Student-concept connection space (matrix).

(b) The number of entries in different intervals corresponding to value of entries.

Fig. 3. (a) A brief example of student-concept connection space on ASSIST09. The value in each box indicates the number of questions answered by a student on a specific concept. (b) The data distribution in student-concept connection space.

best of our knowledge, DKT is the first attempt to leverage recurrent neural networks to model student's knowledge state and estimates the probability of students correctly answering questions according to their knowledge states. Following DKT, plentiful works have emerged. DKT+ is an extended variant of DKT, which aims at addressing two major problems in original DKT model by introducing the regularization on prediction consistency. Different from DKT, DHKT (Wang, Ma, & Gao, 2019) utilizes the hierarchical structure between questions and concepts for better prediction. SQKT (Xie, Wang, Song, & Lin, 2021) proposes a question method can capture the latent relationship of questions and design the student attention mechanism to get prediction. EERNNA (Liu et al., 2021b) not only considers the current knowledge state and question, but also takes previous knowledge states into account for prediction. Moreover, there exists some works modelling students' knowledge states by maintaining multiple vectors. For instance, DKVMN (Zhang, Shi, King, & Yeung, 2017) uses a key-value memory network to store concept information and students' knowledge states information. Meanwhile, it makes prediction based on all the vectors of knowledge states. SKVMN (Abdelrahman & Wang, 2019) improves DKVMN by using a modified Long Short-Term Memory (LSTM) with hops, called Hop-LSTM, to better discover the correlation between latent concepts and questions for prediction. Besides, some attention-based methods are proposed. Concretely, SAKT (Pandey & Karypis, 2019) presents a self-attention mechanism for knowledge tracing to capture the relation between students' historical interactions. SAKT has been shown to be far more efficient than the RNN-based KT model. AKT (Ghosh, Heffernan, & Lan, 2020) couples flexible attention-based neural network models and attention weights are computed by a distance-aware exponential decay with a global decay rate. SAINT (Choi et al., 2020b) is the first KT method using an encoder-decoder model that applies deep self-attentive layers to questions and responses separately to better predict students' performance.

Summary. These aforementioned methods make full use of the different networks or attention mechanisms to accurately model the mastery level of students, but they neglect the individualization of questions. In this paper, we strive to predict students' performance by leveraging the fine information encapsulated in interactions between student-question answer space and question-concept association space and inherent factors of each question.

2.2. *Question representation*

Question representation in KT plays a crucial role in predict student's performance on future questions. Normally, question representation methods can be summarized into three categories: leveraging the index information, utilizing the textual information and excavating the relations between questions and concepts. To be specific, the methods, such as DKT (Piech et al., 2015) and DKT+ (Yeung & Yeung, 2018), directly encode the question representation by only combining the index information of question and student's response on this question. Moreover, analogous to DIMKT (Shen et al., 2022) and CL4KT (Lee et al., 2022), this family of methods obtain personalized representation of each question from a created embedding matrix by using the index information. Besides, some methods are proposed to exploit the textual information. For example, RKT (Pandey & Srivastava, 2020) and EKT (Liu et al., 2021b) make full use of the textual context of question to enhance

the question representation.

However, it is troublesome to acquire the text information in some scenarios. In order to get rid of this limitation, PEBG (Liu et al., 2020) maximally extract the underlying relations among questions and concepts in the defined bipartite graph to attain the question representation. Similarly, PERM (Wang et al., 2022b) introduces fine-grained semantic information from student-question-concept global relation to enrich question representation.

Summary. These above mentioned methods either directly leverage the question-related information or fully excavate the implicit information contained in the student-question-concept interactions to generate question representation. Nevertheless, they fail to explore the fine-grained response information of students. Hence, different from the previous work, the fine-grained question representations are deduced from two distinct (incorrect and correct) responses and connections in student-question-concept in this paper.

3. Preliminaries

In this section, we briefly introduce the task of KT with some notations used in this paper.

In KT task, given the set of N students $S = \{s_1, s_2, \ldots, s_i, \ldots, s_N\}$, the set of M questions $Q = \{q_1, q_2, \ldots, q_j, \ldots, q_M\}$ and the set of K concepts $C = \{c_1, c_2, \ldots, c_k, \ldots, c_K\}$. Suppose the learning history of a student is $X = \{(q_1, r_1), (q_2, r_2), \ldots, (q_{t-1}, r_{t-1})\}$, where q_{t-1} denotes the question he answers at timestep $t-1$ and $r_{t-1} \in \{0, 1\}$ is the response result. Generally, if he correctly answers question q_{t-1}, r_{t-1} equals to 1, otherwise r_{t-1} equals to 0. Based on the learning history X, our goal is to predict the probability of correctly answering a new question at timestep t by the student, i.e., $\hat{r}_t = p(r_t = 1 | q_t, X)$. The key mathematical notations used in this paper are summarized in Table 1.

To obtain the student's performance on next question q_t, we define the following terminologies:

Definition 1. (Student-question answer space). We use $\mathbf{R} \in \{-1, 0, 1\}^{N \times M}$ to denote the student-question answer space indicating the students' response data on questions, where $\mathbf{R}_{ij} = -1/1$ if student s_i gives incorrect/correct response to question q_j, and $\mathbf{R}_{ij} = 0$ if student s_i does not give response to question q_j.

Definition 2. (Question-concept association space). We use $\mathbf{Q} \in \{0, 1\}^{M \times K}$ (*i.e.*, a binary question-concept relation matrix labelled by education experts) to denote question-concept association space, indicating whether the question involves a particular concept. If the question q_j involves the concept c_k, $\mathbf{Q}_{jk} = 1$, otherwise $\mathbf{Q}_{jk} = 0$.

Table 1
The key notations and descriptions.

Notation	Description
S	A set of students; $\|S\| = N$; s_i is the ith student
Q	A set of questions; $\|Q\| = M$; q_j is the jth question
C	A set of concepts; $\|C\| = K$; c_k is the kth concept
X	The students' learning history
$\mathbf{Q}$	The question-concept association space
z	z $= -/+$ matches incorrect/correct responses
$\mathbf{R}^{\mathrm{z}}$	The student-question answer spaces with two responses
$\mathbf{A}^{\mathrm{z}}$	The student-concept connection space with two responses
$\mathbf{A}^{\mathrm{HR^z}}$	The hard reliability of student-concept connection space with two responses
$\mathbf{A}^{\mathrm{SSR^z}}$,$\mathbf{A}^{\mathrm{CSR^z}}$	The soft reliability of student-concept connection space under the perspective of students and concepts with two responses
$\mathbf{A}^{\mathrm{SVD^z}}$	The student-concept connection space via reliability calibration and dimensionality reduction with two responses
$\mathbf{m}^{\mathrm{z}}_{t-1}$	The incorrect or correct response representation of q_{t-1}
$\mathbf{h}_t$	The student's knowledge state at timestep t
$\mathbf{q}_t^{\mathrm{DIFF}}$,$q_t^{\mathrm{DISC}}$	The difficulty level and discrimination of q_t
$\mathbf{x}_t$	The ultimate knowledge state at timestep t
r_t, $\hat{r}_t$	The actual response and predicted probability at timestep t

Definition 3. (Knowledge state). The student's knowledge state is his mastery level on K concepts, and the knowledge state of a student at timestep t is represented as $\mathbf{h}_t \in \mathbb{R}^K$.

Definition 4. (Inherent factors of question). The inherent factors of question consist of two parts, that is, question difficulty level and question discrimination. For a concrete question q_j, the former hints at the difficulty level of each concept examined by the question and the latter indicates the capability of question to differentiate between those whose mastery level is high from those with low mastery level, denoted as $\mathbf{q}_j^{\mathrm{DIFF}}$ and q_j^{DISC}, respectively.

4. Method

In this section, we propose a novel multiple question factors for knowledge tracing (MQFKT) method that makes full use of various question factors for better prediction as suggested in Fig. 4. And the details of MQFKT are introduced as follows.

4.1. Response representation module

In the following subsections, we first integrate student-question answer space with question-concept association space to obtain student-concept connection spaces under two partitions. Then, we devise two calibration strategies (*Hard Reliability* as well as *Soft Reliability*) and perform dimensionality reduction on two student-concept connection spaces to acquire their low-dimensional representations. Lastly, the fine response representations on each question are obtained by the average pooling operation of embeddings of concepts that this question requires. The following parts will thoroughly describe how to achieve fine-grained response representations.

4.1.1. Student-concept connection space construction

We first divide the student-question answer space with respect to incorrect and correct responses and further denote two student-question answer spaces as $\mathbf{R}^{\mathrm{z}}$ with z $\in \{-, +\}$ ($-$ matches incorrect responses and $+$ corresponds correct responses), where $\mathbf{R}^{\mathrm{z}}$ has the same dimension as $\mathbf{R}$.

After that, we simply integrate student-question answer space $\mathbf{R}^{\mathrm{z}}$ and question-concept association space $\mathbf{Q}$ to establish the student-concept connection space about two responses with the following form:

$$\mathbf{A}^{\mathrm{z}} = \mathbf{R}^{\mathrm{z}}\mathbf{Q} \tag{1}$$

where we have $\mathbf{A}^{\mathrm{z}} \in \mathbb{R}^{N \times K}$. A concrete entry in $\mathbf{A}^{\mathrm{z}}$ such as $\mathbf{A}^{\mathrm{z}}_{ik}$ represents the number of questions related to concept c_k answered by student s_i.

4.1.2. Reliability calibration

Intuitively, $\mathbf{A}^{\mathrm{z}}_{ik}$ can be roughly approximated as the proficiency level of student s_i on the particular concept c_k. Therefore, a large value of entry indicates its strong reliability while a little value means it is less reliable. To rationally quantify the entries' reliability and eliminate their bias under various numerical scales, we devise the *Hard Reliability* and *Soft Reliability* strategies from global and local perspectives, respectively. Specifically, from the global student-concept connection space's perspective, *Hard Reliability* directly compares the absolute magnitude between the value of a specific entry (i.e., $\mathbf{A}^{\mathrm{z}}_{ik}$) and the value of remaining entries in this space. *Soft Reliability* finely focuses on the particular student and concept. Along this line, from the local student and concept perspective, it considers separately the relative magnitude of $\mathbf{A}^{\mathrm{z}}_{ik}$ in its row (i.e., student s_i) and column (i.e., concept c_k). Next, we will illustrate these two strategies in detail.

Hard Reliability. Since entries of the student-concept connection space under two responses are various but relatively concentrated as illustrated in Fig. 3(b), we intuitively argue that the larger the value of entry, the more reliable it is. Specifically, for student-concept

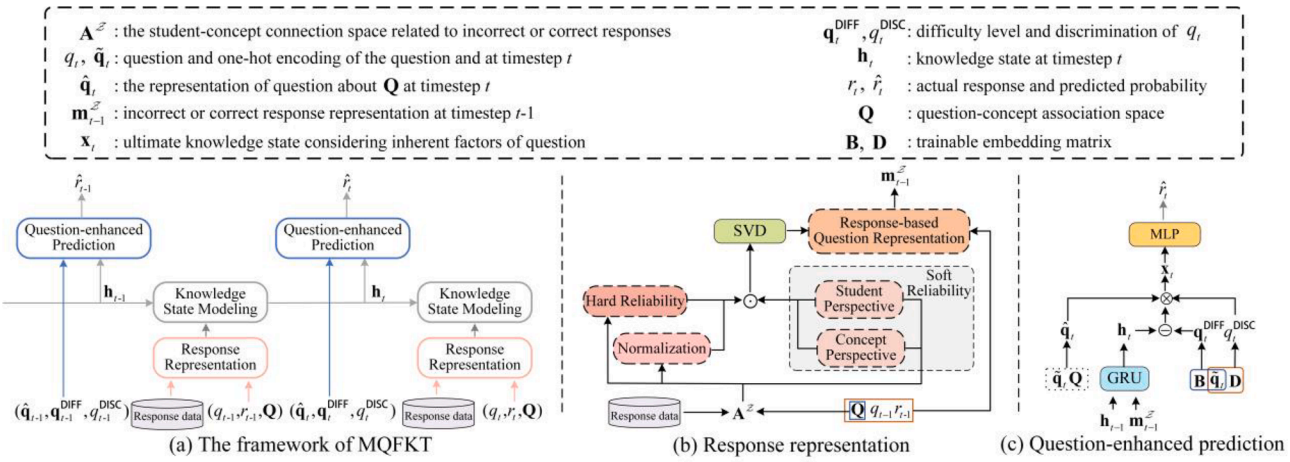


Fig. 4. The frameworks and details of MQFKT. (a) The overview of MQFKT mainly contains three modules: response representation module, knowledge state modeling module and question-enhanced prediction module. (b) The details of response representation module illustrate how to obtain the fine response representations under incorrect and correct responses to each question via sufficiently excavating and differentiating the students' responses. (c) The details of question-enhanced prediction module primarily show how to predict student's performance with the ultimate knowledge state combining student's knowledge state and inherent factors of question.

connection space $\mathbf{A}^{\mathcal{Z}}$, its reliability $\mathbf{A}^{\mathrm{HR}^{\mathcal{Z}}}_{ik}$ corresponding to $\mathbf{A}^{\mathcal{Z}}_{ik}$, namely hard reliability, is calculated as:

$$\mathbf{A}^{\mathrm{HR}^{\mathcal{Z}}}_{ik} = \sigma\left(\mathbf{A}^{\mathcal{Z}}_{ik}\right) \tag{2}$$

where $\sigma(\cdot)$ is the *sigmoid function*. Along this line, hard reliability of each entry can be obtained, eventually represented as $\mathbf{A}^{\mathrm{HR}^{\mathcal{Z}}}$.

Soft Reliability. We design the soft reliability from two perspectives (i.e., students and concepts). Specifically, for one thing, from the perspective of students, the number of question responses involving a specific concept varies from student to student, and our intuition is that the higher the number, the more reliable it is. And for another, from the perspective of concepts, a certain student gives different number of responses to questions that touch on different concepts, and it is intuitive that the bigger the number, the more reliable it is. Based on the above two considerations, for student-concept connection space related to incorrect and correct responses, we first separately obtain average number of questions $\overline{s}^{\mathcal{Z}}_k$ including the concept c_k and answered by students under the perspective of students, and $\overline{c}^{\mathcal{Z}}_i$ containing concepts that the student s_i has interacted with under the perspective of concepts as follows:

$$\overline{s}^{\mathcal{Z}}_k = \frac{\sum_{i=1}^{N} \mathbf{A}^{\mathcal{Z}}_{ik}}{\sum_{i=1}^{N} \mathrm{II}\left(\mathbf{A}^{\mathcal{Z}}_{ik} \neq 0\right)} \tag{3}$$

$$\overline{c}^{\mathcal{Z}}_i = \frac{\sum_{k=1}^{K} \mathbf{A}^{\mathcal{Z}}_{ik}}{\sum_{k=1}^{K} \mathrm{II}\left(\mathbf{A}^{\mathcal{Z}}_{ik} \neq 0\right)} \tag{4}$$

where $\mathrm{II}(\cdot)$ denotes the indicator function.

Subsequently, two soft reliabilities $\mathbf{A}^{\mathrm{SSR}^{\mathcal{Z}}}_{ik}$ and $\mathbf{A}^{\mathrm{CSR}^{\mathcal{Z}}}_{ik}$ corresponding to $\mathbf{A}^{\mathcal{Z}}_{ik}$ can be acquired under the perspective of students and concepts as:

$$\mathbf{A}^{\mathrm{SSR}^{\mathcal{Z}}}_{ik} = \begin{cases} 1, & \mathbf{A}^{\mathcal{Z}}_{ik} > \overline{s}^{\mathcal{Z}}_k \\ 1 - \exp\left(-\mathbf{A}^{\mathcal{Z}}_{ik}\right), & otherwise \end{cases} \tag{5}$$

$$\mathbf{A}^{\mathrm{CSR}^{\mathcal{Z}}}_{ik} = \begin{cases} 1, & \mathbf{A}^{\mathcal{Z}}_{ik} > \overline{c}^{\mathcal{Z}}_i \\ 1 - \exp\left(-\mathbf{A}^{\mathcal{Z}}_{ik}\right), & otherwise \end{cases} \tag{6}$$

where the reason for choosing $y = 1 - \exp(-x)$ as soft reliability calibration function is that it can modify the entries which have a considerable proportion in student-concept connection space, as the first interval in Fig. 3(b) shows. Then, we can obtain $\mathbf{A}^{\mathrm{SSR}^{\mathcal{Z}}}$ and $\mathbf{A}^{\mathrm{CSR}^{\mathcal{Z}}}$.

The calibrated student-concept connection space $\mathbf{A}^{\mathrm{NEW}^{\mathcal{Z}}} \in \mathbb{R}^{N \times K}$ related to two responses can be updated eventually:

$$\mathbf{A}^{\mathrm{NEW}^{\mathcal{Z}}} = \mathbf{A}^{\mathrm{NOR}^{\mathcal{Z}}} \odot \mathbf{A}^{\mathrm{HR}^{\mathcal{Z}}} \odot \mathbf{A}^{\mathrm{SSR}^{\mathcal{Z}}} \odot \mathbf{A}^{\mathrm{CSR}^{\mathcal{Z}}} \tag{7}$$

where $\mathbf{A}^{\mathrm{NOR}} = \mathrm{normal}\ (\mathbf{A})$ and $\mathrm{normal}\ (\cdot)$ denotes operations to normalize rows and columns of matrix with *softmax function*. $\odot$ denotes the Hadamard product. Up to now, we establish the calibrated student-concept connection space based on incorrect and correct responses considering the reliability of students' cognitive state.

4.1.3. Dimensionality reduction

Next, owing to the fact that the sparsity of both question-concept association space and student-question answer space leads to the sparseness of the final student-concept connection space. We thus decompose it using SVD (Huo et al., 2020) as follows:

$$\mathbf{A}^{\mathrm{NEW}^{\mathcal{Z}}} = \mathbf{N}\mathbf{\Sigma}\mathbf{K}^{T} \tag{8}$$

where $\mathbf{N} \in \mathbb{R}^{N \times N}$ is the decomposed student matrix, $\mathbf{\Sigma} \in \mathbb{R}^{N \times K}$ is the diagonal value matrix, and $\mathbf{K} \in \mathbb{R}^{K \times K}$ is the decomposed concept matrix. We select the N' number of diagonal values to approximate $\mathbf{A}^{\mathrm{NEW}^{\mathcal{Z}}}$. Afterwards, we have $\mathbf{N}' \in \mathbb{R}^{N' \times N}$. The low-dimensional representations of student-concept connection space $\mathbf{A}^{\mathrm{SVD}^{\mathcal{Z}}}$ associated with incorrect and correct responses can be calculated as:

$$\mathbf{A}^{\mathrm{SVD}^{\mathcal{Z}}} = \mathbf{N}' \mathbf{A}^{\mathrm{NEW}^{\mathcal{Z}}} \tag{9}$$

where $\mathbf{A}^{\mathrm{SVD}^{\mathcal{Z}}} \in \mathbb{R}^{N' \times K}$. Each column in $\mathbf{A}^{\mathrm{SVD}^{\mathcal{Z}}}$ is a low-dimensional representation of the corresponding concept. The main purpose of introducing SVD is that it has an excellent power of low-rank representation and the ability to extract global student information, so as to obtain the perfect low-dimensional representation of student-concept connection space.

4.1.4. Response-based question representation

Since the response information of students on the questions plays an indispensable role in updating their knowledge states at the next timestep. Consequently, we assume that students give different responses on the question q_{t-1} at timestep t-1 and its corresponding fine representation is denoted as $\mathbf{m}^{\mathcal{Z}}_{t-1} \in \mathbb{R}^{N}$. Next, we use the following formulation to obtain the incorrect or correct response representation on the basis of the concepts corresponding to q_{t-1}:

$$\mathbf{m}_{t-1}^{z} = \frac{1}{|C_{t-1}|} \sum_{c_k \in C_{t-1}} \left(\mathbf{A}^{\mathrm{SVD}^z} \widetilde{\mathbf{c}}_k\right) \tag{10}$$

where C_{t-1} represents a set of concepts related to the question q_{t-1}. $\widetilde{\mathbf{c}}_k$ is one-hot encoding of the concept c_k. It is worth noting that the above obtained response representation of the question is not only low in dimension, but also fine, avoiding properly the curse of dimensionality and enormous sparsity and excavating adequately the question information contained in interactions between student-question answer space and question-concept association space.

4.2. Knowledge state modeling module

Subsequently, we feed fine response representation to the gated Recurrent Neural Networks to update the student's knowledge state. It is worth noting that the Gated Recurrent Unit (GRU) has higher computational efficiency and simpler structure compared with Long Short Term Memory (LSTM), but the performance of LSTM is on a par with GRU. Therefore, we here adopt the GRU to model knowledge state about student:

$$\mathbf{h}_t = \mathrm{GRU}\left(\mathbf{m}_{t-1}^{z},\ \mathbf{h}_{t-1}\right) \tag{11}$$

Note that knowledge state modeling module in KT is designed distinctly in most existing studies. We argue it is promising to replace the knowledge state modeling module with existing KT methods, which may lead to better performance. Here, we omit it for brevity, as it is not our main concern and leave this part for the future.

4.3. Question-enhanced prediction module

It is worthy of finding ways to consider factors related to questions for better prediction. In the follow-up parts, we introduce how to project questions to factor vectors and model the complex interactions with the difficulty level of each question.

4.3.1. Question-enhanced cognition estimation

Specifically, we directly obtain the concept vector $\widehat{\mathbf{q}}_t$ corresponding to concepts contained in question at timestep t on the basis of the pre-given question-concept association space $\mathbf{Q}$:

$$\widehat{\mathbf{q}}_t = \mathbf{Q}^T \widetilde{\mathbf{q}}_t \tag{12}$$

where $\widehat{\mathbf{q}}_t \in \mathbb{R}^K$, and $\widetilde{\mathbf{q}}_t$ is the one-hot encoding of the question. Next, we introduce the inherent factors of question: difficulty level $\mathbf{q}_t^{\mathrm{DIFF}}$ and discrimination q_t^{DISC} of question. They can be obtained by:

$$\mathbf{q}_t^{\mathrm{DIFF}} = \sigma(\mathbf{B}\widetilde{\mathbf{q}}_t), q_t^{\mathrm{DISC}} = \sigma\left(\mathbf{D}^T \widetilde{\mathbf{q}}_t\right) \tag{13}$$

where $\mathbf{B} \in \mathbb{R}^{K \times M}$ and $\mathbf{D} \in \mathbb{R}^M$ are trainable parameters. Up to now, we project questions to different factor vectors.

4.3.2. Response prediction

After obtaining the factor vectors about questions, we design the neural layers for modeling interactions between students and questions inspired by Wang et al. (2020) and Converse et al. (2021). We formulate the first layer of the interaction layers as:

$$\mathbf{x}_t = \widehat{\mathbf{q}}_t^{\ \circ}\left(\mathbf{h}_t - \mathbf{q}_t^{\mathrm{DIFF}}\right) \times q_t^{\mathrm{DISC}} \tag{14}$$

where $\mathbf{x}_t$ represents the ultimate knowledge state taking into account question factors and $\circ$ means element-wise product between vectors. The root of modeling interaction layers comes from MIRT (Reckase, 2009) that is a classical educational psychology theory. It has the convincing hypothesis that the mastery level of student is determined by the combination of student and question related factors. Afterwards, we feed $\mathbf{x}_t$ to MLP with three layers to obtain the probability $\widehat{r}_t$ of correctly answering the new question q_t:

$$\widehat{r}_t = \mathrm{MLP}(\mathbf{x}_t) \tag{15}$$

It deserves to mention that we restrict each element of weight matrix of MLP to be positive to satisfy the monotonicity assumption that plays a crucial role in ensuring the interpretation of interactions.

To optimize the model, we use a common cross-entropy loss between the predicted probability of being correctly answered and the true label. The following objective function is defined over training data:

$$L = -\sum_t (r_t \log \widehat{r}_t + (1 - r_t)\log(1 - \widehat{r}_t)) \tag{16}$$

The Algorithm 1 sums up our MQFKT method.

4.4. Model discussion

In this subsection, we conduct some interesting analysis to better illustrate MQFKT method. In this subsection, we conduct deep analysis to better illustrate the MQFKT method. And we first summarize the advantages of our method from two aspects: (1) Response representation. MQFKT sufficiently exploits the students' response information on questions to generate the fine-grained response representation, which is different from existing studies, such as PEBG and SQKT; (2) Question-enhanced prediction. MQFKT leverages the complex interaction considering the difficulty level and discrimination of questions to effectively model the relationship between students and questions by following the common paradigm, such as NeuralCD (Wang et al., 2020), RCD (Gao et al., 2021) and CT-NCM (Ma et al., 2022), so as to achieve the greater prediction performance.

Along this line, we choose two typical studies (i.e., PEBG and NeuralCD) based on the above two aspects for discussing the relation between our method and these typical studies.

4.4.1. Relation with PEBG

PEBG proposes a pre-training method to obtain the question representation with the help of question difficulties and three types of relations (i.e., explicit question-concept relations, concept similarity, and question similarity). MQFKT is distinguished from two aspects: (1) PEBG first constructs the question-concept relations as a bipartite graph to learn low-dimensional question representations for knowledge tracing. However, MQFKT first exploits the response information contained in the student-question-concept connections to generate low-dimensional incorrect and correct response representations to questions; (2) PEBG utilizes matrix factorization to reduce the complexity of representation. In MQFKT, SVD is employed to reduce it.

4.4.2. Relation with NeuralCD

NeuralCD obtains the projected student and question factors and further sufficiently captures the complex interaction relations between students and questions with DNNs. MQFKT also uses these interaction relations but distinguishes from NeuralCD in the following aspects: (1) NeuralCD is normally used to estimate the students' proficiency level on particular concepts under static cognitive diagnosis scenarios. Nevertheless, MQFKT aims to predict students' performance on future questions in dynamic knowledge tracing scenarios; (2) The student-related input of interactive function in NeuralCD is acquired from a trainable matrix. However, the student-related input in MQFKT is generated by means of deeply excavating two types of response information.

5. Experiments

In this section, we conduct extensive experiments to evaluate the performance of our proposed MQFKT framework. To be specific, we first introduce the datasets and describe our experiment settings in detail. Then we also make some discussions with an extended investigation to demonstrate the effectiveness of our method by answering the following

research questions:

- **RQ1**: How does our proposed MQFKT perform in accuracy compared to the state-of-the-art KT methods?
- **RQ2**: What is the influence of crucial components of MQFKT, including reliability calibration, dimensionality reduction, and question-enhanced prediction module?
- **RQ3**: Can MQFKT generate meaningful response representations of questions?
- **RQ4**: How about the interpretation of question difficulty recovered by MQFKT method?

5.1. Datasets

We validate the effectiveness of our model on three public real-world datasets: ASSIST09[1], ASSISTchall[2], and EdNet-KT1[3].

ASSIST09 which is widely used by researchers for purposes such as student modeling is collected from the free online educational ASSISTment platform. Despite the popularity of this dataset, its original version is not reliable as discussed in Zhang et al. (2017). We thus choose the corrected version fixing the data modeling issues (Wang et al., 2021).

Algorithm 1 The MQFKT method for predicting students' performance

Input: The set of students S, the set of questions Q, the set of concepts C, the learning history X, and the question-concept association space $\mathbf{Q}$.
Output: the predicted probability $\hat{r}_t = p(r_t = 1|q_t, X)$.
1: Partition the student-question answer space based the response data;
2: Obtain the student-concept connection space $\mathbf{A}^z$ corresponding to incorrect or correct response by combining $\mathbf{Q}$ and Eq. (1);
3: for $i = 1$ to N
4: for $k = 1$ to K
5: Calculate the hard reliability $\mathbf{A}_{ik}^{\mathrm{HR}^z}$ according to Eq. (2);
6: Calculate the soft reliability $\mathbf{A}_{ik}^{\mathrm{SSR}^z}$ from the view of students and $\mathbf{A}_{ik}^{\mathrm{CSR}^z}$ from the view of concepts according to Eq. (5) and Eq. (6);
7: Calculate the normalized entry $\mathbf{A}_{ik}^{\mathrm{NOR}^z}$ corresponding to $\mathbf{A}_{ik}^z$;
8: Compute the calibrated entry $\mathbf{A}_{ik}^{\mathrm{NEW}^z}$ via step 5 to step 7 according to Eq. (7);
9: end for;
10: end for;
11: Attain the low-dimensional student-concept connection space $\mathbf{A}^{\mathrm{SVD}^z}$ by Eq. (8) and Eq. (9);
12: Obtain the fine question representation $\mathbf{m}_{t-1}^z$ according to Eq. (10);
13: Capture knowledge state $\mathbf{h}_t$ according to Eq. (11);
14: Project the questions to two factor vectors: $\mathbf{q}_t^{\mathrm{DIFF}}$ related to difficulty level q_t^{DISC} related to discrimination according to Eq. (13);
15: Obtain the ultimate mastery vector $\mathbf{x}_t$ considering difficulty level and discrimination according to Eq. (14);
16: Predict the performance $\hat{r}_t$ on future question q_t according to Eq. (15);
17: Update the parameters of model with backpropagation algorithm until convergence according to Eq. (16);

ASSISTchall is obtained from 2017 ASSISTment data mining competition. It is gathered from a longitudinal study that tracks students from their use of ASSISTment.

EdNet-KT1, the largest publicly available interaction dataset, is collected from the intelligent online tutoring platform named Santa dedicated to practicing English for international communication (TOEIC) assessment (Choi et al., 2020a). We reduce the size of EdNet-KT1 dataset by randomly sampling 5000 users among 600K users.

Note that for dataset pre-processing, we filter out students with less than 15 response logs for ASSIST09 and ASSISTchall to guarantee that each student has enough data for knowledge tracing. Moreover, we only select students with more than 10 response logs for EdNet-KT1. The detailed statistics of three datasets after pre-processing are shown in Table 2.

Table 2
Dataset summary.

Summary	ASSIST09	ASSISTchall	EdNet-KT1
#Students	4163	1709	5000
#Questions	17,746	2210	12,081
#Concepts	123	101	189
#Response records	278,868	351,820	632,230
Rate of incorrect response	34.25%	18.44%	31.59%

5.2. Experimental setup

5.2.1. Implementation settings

We set N' in dimensionality reduction that enables the MQFKT to achieve best performance in Section 5.3, and the size of hidden states as 64. Moreover, the dimensions of the MLP with three layers in Eq. (15) are 512, 256 and 1 respectively. Besides, we initialize the whole parameters with Xavier initialization, which fill the weights with random values sampled from $N \sim (0, std^2)$, where $std = \sqrt{2/(n_{in} + n_{out})}$. n_{in} is the number of neurons feeding into the weights, and n_{out} is the number of neurons the result is fed to. For each dataset, we conduct 5-fold cross-validation for all models. Specifically, for each fold, we split 80% of the learning history as the training set as well as validation set (their ratio is 8:2), and the rest as the testing set. Meanwhile, we use the Adam algorithm (Kingma & Ba, 2014) for optimization with an initial learning rate of 0.001 and a batch size of 64. Our method is implemented by PyTorch using Python, and all experiments are run on a Linux server with Quadro RTX 6000 GPU with 24G of video memory.

For the baselines which use RNN to update the knowledge states such as DKT, DKT + and EERNNA, we also apply GRU for fair comparison. In order to make the results comparable, the other hyperparameters of all the baselines are carefully tuned to their best performance.

5.2.2. Baselines and evaluation metrics

To evaluate the effectiveness of our proposed model, we use some KT models as our baselines. Specifically, we applied it on four kinds of KT methods: traditional methods (BKT, KTM) that utilize the factors associated with students' learning to predict students' responses, attention-based methods (SAKT, AKT-NR, SAINT) that monitor knowledge mastery of students according to attention mechanism, multi-state methods (DKVMN, SKVMN) that maintain multiple vectors to represent students' knowledge states and single-state methods (DKT, DKT+, DHKT, SQKT, EERNNA) maintaining one vector to represent the knowledge state of a student.

(1) *BKT*: This is a classical knowledge tracing method using Hidden Markov Model which hypothesizes student's knowledge state as a set of binary variables and update them on the basis of Bayesian rules. (e.g., Corbett & Anderson, 1995)
(2) *KTM*: This is the traditional factor analysis method based on Factorization Machine, which can use many types of features like questions, concepts to predict student's responses. In this paper, for fairness we only consider questions and concepts as its side information in comparison. (e.g., Vie & Kashima, 2019)
(3) *SAKT*: SAKT proposes a self-attention mechanism for knowledge tracing to capture the relation between student's historical interactions. SAKT has been shown to be far more efficient than the RNN-based KT model. (e.g., Pandey & Karypis, 2019)
(4) *AKT*: AKT exploits flexible monotonic attention-based neural network models and attention weights are computed by a distance-aware exponential decay with a global decay rate. Moreover, the Rasch model is introduced to generate the individual concept and question embeddings. (e.g., Ghosh, Heffernan, & Lan, 2020)

[1] https://sites.google.com/site/assistmentsdata/home/assistment-2009–2010-data.
[2] https://sites.google.com/view/assistmentsdatamining/dataset.
[3] https://ednet-leaderboard.s3-website-ap-northeast-1.amazonaws.com/.

(5) *SAINT*: This is the first KT method using an encoder-decoder model that applies deep self-attentive layers to questions and responses separately to better predict student's performance. (e.g., Choi et al., 2020b)
(6) *DKVMN*: DKVMN takes advantage of memory augmented neural network to track student's knowledge state. It uses a key matrix and a value matrix to respectively store underlying concept representation and update the state. (e.g., Zhang et al., 2017)
(7) *SKVMN*: SKVMN improves DKVMN by using a modified LSTM with hops, called Hop-LSTM, to better discover the correlation between latent concepts and questions for prediction. (e.g., Abdelrahman & Wang, 2019)
(8) *DKT*: This is a seminal KT method that leverages a LSTM to model student's knowledge state and trains a unified network to predict student's response on next question. The inputs are the one-hot encodings of question. (e.g., Piech et al., 2015)
(9) *DKT+*: This is an extended variant of DKT, which aims at addressing two major problems in original DKT model by introducing the regularization on prediction consistency. Specifically, the first problem is that DKT model fails to the observed input, and the second problem is the predicted performance for DKT across timesteps is not consistent. (e.g., Yeung & Yeung, 2018)
(10) *DHKT*: DHKT incorporates the hierarchical information between questions and concepts into knowledge tracing model and effectively obtains meaningful representations of questions. (Wang, Ma, & Gao, 2019)
(11) *SQKT*: SQKT devises the question representation method based on the designed weighted graph and the correlated concepts. Additionally, the designed student attention mechanism is dedicated to updating students' knowledge state to achieve higher prediction accuracy. (Xie, Wang, Song, & Lin, 2021)
(12) *EERNNA*: This is an improved version of DKT that not only considers the current knowledge state and question, but also takes previous knowledge states into account for prediction. (e.g., Liu et al., 2021b)

Moreover, we evaluate the models on these datasets from classification perspective. During evaluation, we argue that students' response result answered incorrectly or correctly can be separately represented as 0 and 1. Thereby, we conduct the *t*-test under Accuracy (ACC, the higher the better) metric and Area Under the ROC Curve (AUC, the higher the better) metric for model evaluation. Note that p-value < 0.05 implies that the method is statistically significant.

5.3. Performance comparison (RQ1)

To answer the RQ1, in this part, we compare the performance of MQFKT with baseline methods. Table 3 shows the overall evaluation results about the student performance prediction task, where we conduct 5-fold cross validation and report the average results. The best performance and the second-best performance models are denoted in bold and underlined, respectively. The results in Table 3 suggest some interesting conclusions:

Firstly, we can easily see that traditional methods are inferior to other methods in most cases, which seems to be since they have limitations to capture the sequential and temporal characteristics of students' learning history. Secondly, The SQKT has better performance than EERNNA on EdNet-KT1, which reveals that introducing student attention mechanism into LSTM is more effective to promote the prediction on larger datasets. Thirdly, we notice that MQFKT significantly outperforms state-of-the-art EERNNA method on ASSIST09 and EdNet-KT1, which improves the AUC by 3.6% and 2.7% respectively. However, it has the least improvement on ASSISTchall, which purely promotes the AUC by 1.2%. We argue the reason is that the more information of incorrect responses is available (i.e., the higher rate of incorrect response), the better it is for prediction. Last but not least, MQFKT performs consistently better than them in terms of both ACC and AUC on all datasets compared to other baselines, which indicates that exploiting the information of responses about questions to obtain fine response representations of questions and integrating the difficulty level of each question into prediction are essential and necessary to get better performance. Up to now, RQ1 has been answered in detail.

5.4. Ablation experiments (RQ2)

To answer the RQ2, we conduct an ablation study in terms of AUC and ACC on all three real-world datasets to further verify the effectiveness of each component in our method. We set three comparative settings, and the details of the three settings are listed below:

- MQFKT-RC refers to MQFKT without reliability calibration, i.e., we do not modify unreliable values of entries in student-concept connection spaces corresponding to incorrect and correct response.
- MQFKT-DR refers to MQFKT that does not utilize the way of dimensionality reduction to obtain the low-dimensional response representation of the question.
- MQFKT-QPM refers to MQFKT without question-enhanced prediction module, where we directly leverage the students' knowledge state acquired through GRU to predict their performance on future questions.

To be fair, except for the changes mentioned above, the other parts of aforementioned variant methods and implementation settings remain identical. From Fig. 5, we first can observe that dimensionality reduction component has a minimal contribution to the performance in the whole variants but is harmful to accurately model the evolution of the

Table 3
Comparison results on student performance prediction on three datasets. * indicates the p-value < 0.05 in the significance test.

Methods		ASSIST09		ASSISTchall		EdNet-KT1	
		ACC	AUC	ACC	AUC	ACC	AUC
traditional	BKT	0.6342	0.6511	0.5874	0.6017	0.6738	0.5819
	KTM	0.7144	0.7124	0.6764	0.6552	0.6634	0.6994
attention-based	SAKT	0.7681	0.7861	0.7144	0.7261	0.7486	0.7665
	AKT-NR	0.7601	0.7641	0.7173	0.7282	0.7246	0.7473
	SAINT	0.7358	0.7441	0.6922	0.7087	0.7118	0.7277
multi-state	DKVMN	0.7112	0.7219	0.6701	0.6774	0.6887	0.7087
	SKVMN	0.7244	0.7441	0.6926	0.7098	0.7066	0.7368
single-state	DKT	0.7362	0.7561	0.7098	0.7178	0.7141	0.7438
	DKT+	0.7514	0.7824	0.7114	0.7207	0.7223	0.7566
	DHKT	0.7623	0.7786	0.7123	0.7221	0.7317	0.7455
	SQKT	0.7708	0.7922	0.7169	0.7280	<u>0.7549</u>	<u>0.7705</u>
	EERNNA	<u>0.7782</u>	<u>0.8024</u>	<u>0.7198</u>	<u>0.7340</u>	0.7511	0.7614
	MQFKT	**0.7975***	**0.8313***	**0.7211***	**0.7426***	**0.7606***	**0.7820***

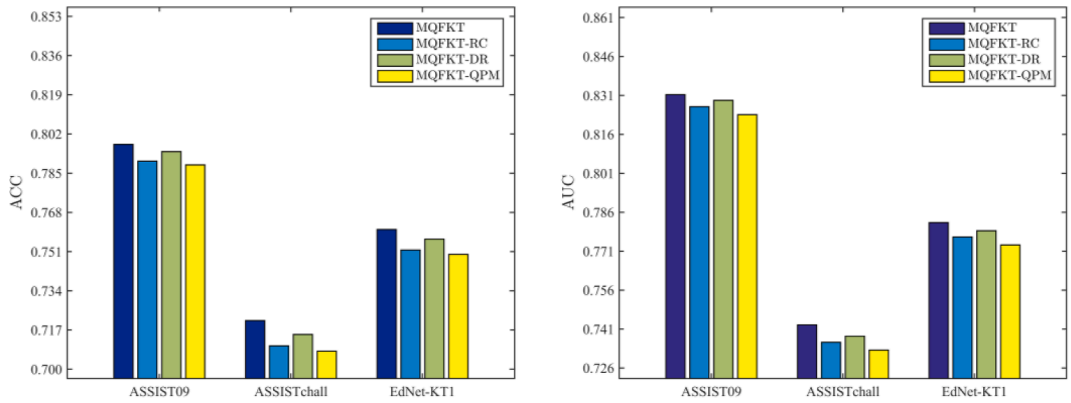


Fig. 5. Performance comparison of ablation study on three real-world datasets.

knowledge states of students if omitted. Second, we can find that question-enhanced prediction module plays a pivotal role in performance prediction, which can cause the biggest decline of predictive results if we do not consider it. That means designing the complex interactions between students and questions including knowledge states can predict better. Lastly, removing the reliability calibration or question-enhanced prediction module has more significant influence on the performance than removing dimensionality reduction, which implies that making full use of question factors (i.e., incorrect responses on questions as well as difficulty level and discrimination differ from question to question) to model students' learning process could achieve an exciting result. So far, we have dealt with RQ2 well.

Moreover, we also devise experiments to compare the results of different number of selected diagonal values N' in dimensionality reduction on student performance prediction. Specifically, we set $N' = \varepsilon \times N$, where ε is selected from {0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1}. Fig. 6 illustrates some interesting comparison results, and we can see that MQFKT achieves the best performance at ε = 0.5, 0.7, 0.4 on ASSIST09, ASSISTchall, and EdNet-KT1 respectively. In contrast, the performance of the method at ε = 1 (i.e., without dimensionality reduction) is relatively mediocre. Besides, both ACC and AUC have better performance in the range of 0.4 to 0.7 for three datasets. These observations demonstrate that dimensionality reduction within a reasonable range is helpful and indispensable for prediction.

5.5. Study of response representation (RQ3)

To answer the RQ3, we intuitively investigate the relationship between these fine response representations since response representation can thoroughly reveal the semantics of response on question. First, we randomly select 300 questions with both incorrect and correct responses on ASSIST09, respectively. Then, we separately visualize incorrect and correct response representations by leveraging the T-SNE method (Van der Maaten & Hinton, 2008). As shown in Fig. 7(a) and Fig. 7(b), those incorrect or correct response representations with same color share same concepts and the clustering results show well meanings. We can observe that both response representations with same color are basically in the same cluster, which is in line with our cognition, i.e., incorrect or correct response representations of questions with the same or similar concepts are of comparable.

Next, we attempt to illustrate the interpretability of fine-grained response representations on questions in MQFKT. Specifically, we have chosen seven questions with six different concepts answered by students repeatedly on ASSIST09, where four questions (i.e., q_{3006}, q_{5464}, q_{6485} and q_{14750}) and three questions (i.e., q_{6527}, q_{8779} and q_{13482}) are either responded correctly or incorrectly, respectively. Afterwards, we visualize the relevant weight between these questions by calculating their cosine similarities as shown in Fig. 8, from which we find some interesting observations. Firstly, relevant weights tend to be higher

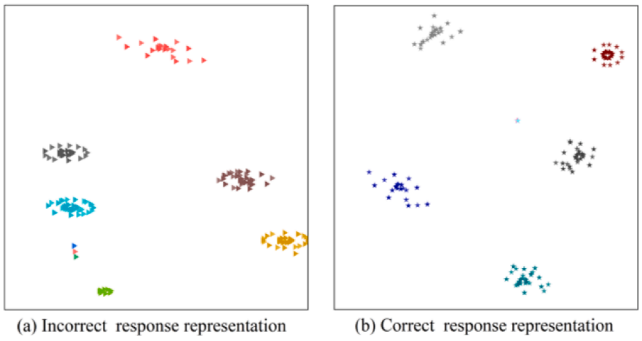


Fig. 7. The visualization of incorrect and correct response representation of chosen questions on ASSIST09. Questions under the same concepts are labelled in the same color.

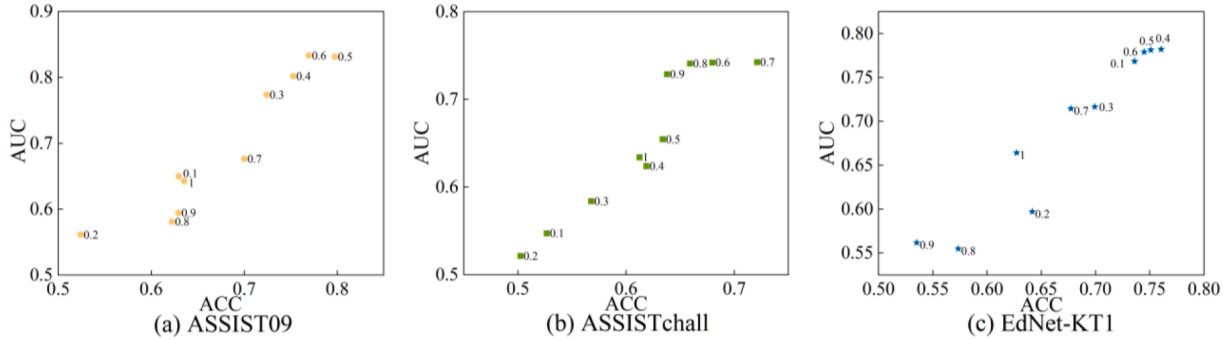


Fig. 6. Performance comparison of different ε achieved by MQFKT on three real-world datasets.

Response Representation	Related Concepts
$\mathbf{m}^{+}_{3006}$	Ordering Fractions
$\mathbf{m}^{+}_{5464}$	Proportion
$\mathbf{m}^{+}_{6485}$	Prime Number
$\mathbf{m}^{+}_{14750}$	Square Root
$\mathbf{m}^{-}_{6527}$	Prime Number
$\mathbf{m}^{-}_{8779}$	Find Pattern
$\mathbf{m}^{-}_{13482}$	Exponents

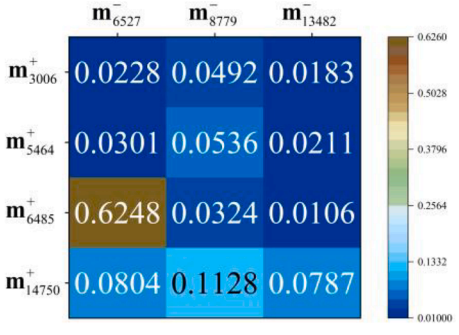


Fig. 8. The seven questions' information and visualization of relevant weights between corresponding response representation on ASSIST09.

between responses associated with the same concept. For instance, response representation of q_{6485} and q_{6527} share the highest weight since they own the same concepts (i.e., *Prime Number*). Furthermore, the relevant weight between q_{14750} and q_{8779} is also high, which can be postulated that q_{14750} is more difficult than q_{8779}. Therefore, the former is answered correctly due to students' guessing while the latter is responded incorrectly because of students' slipping. This results in a smaller difference and a larger correlation weight between corresponding response representations of two questions. This assumption will be verified in Section 5.6. These observations imply that our proposed MQFKT provides a good way to generate fine response representations of questions. Up to this point, we have responded sufficiently to RQ3.

5.6. Prediction process of student's performance (RQ4)

To answer the RQ4, Fig. 9 visualizes the learning sequence and performance prediction of a case student (ID 466) from ASSIST09, where statistical difficulty for each question is computed by the proportion of students that correctly answer this question and prediction difficulty is deduced via MQFKT. We deliberately select this student because he has left correct and incorrect responses on q_{14750} and q_{8779} separately, which are discussed in the previous subsection.

We have the following observations. First, the difference between the predicted difficulty and statistical difficulty is slight in most cases. Nevertheless, the distinction is large for q_{2568} as well as q_{14788}. In addition, the actual response and predicted response of these questions are consistent with prediction difficulty but opposed to statistical difficulty. The reason is that the number of responses on the above questions (i.e., 11 as well as 6) is relatively small, resulting in unreliable statistical difficulty. This suggests that MQFKT can model the difficulty level of questions even the number of responses is limited. Secondly, even the student has answered questions correctly with concept of 48: *Find Pattern* with prediction difficulty of 0.23, 0.27 and 0.11, MQFKT successfully predicts that he would leave the incorrect response on the question with same concept with harder prediction difficulty of 0.45. Thirdly, the predicted difficulty of q_{14750} (i.e., 0.44) with actual response of 1 is obviously higher than that of q_{8779} (i.e., 0.12) with actual response of 0. This indicates that the student gives correct response on q_{14750} because of guessing factor while incorrectly answers q_{8779} with slipping factor, which verifies our hypothesis in Section 5.5.

In a word, the above results demonstrate exploiting individualized difficulty level in our method is of high interpretability. Simultaneously, RQ4 receives an exhaustive reply.

6. Conclusions and future work

In this paper, we present a novel MQFKT method that sufficiently excavating diverse question factors to enhance KT. To be specific, we have deliberately designed a calibrated student-concept connection space and perform dimensionality reduction to acquire the fine response representations. Moreover, difficulty level and discrimination related to questions are properly introduced with neural layers to simulate the complex interactions between students and questions for better prediction. Extensive experiments conducted on three real-world datasets demonstrates the effectiveness and interpretability of MQFKT. However, two main limitations still exist in the proposed method. On one hand, this method solely considers the question factors and fails to take advantage of the student factors. On the other hand, we simply construct interactions between students and questions without incorporating the complex pedagogical and psychological theories to make the model more interpretable and reasonable.

For the future work, we would like to introduce various student factors such as the common phenomenon of forgetting and the answer time to obtain more fine representations of questions. Meanwhile, inspired by the success of contrastive learning, we would try to utilize self-supervised signals in contrastive learning to capture more effective knowledge states, further promoting performance prediction.

Declaration of Competing Interest

The authors declare that they have no known competing financial

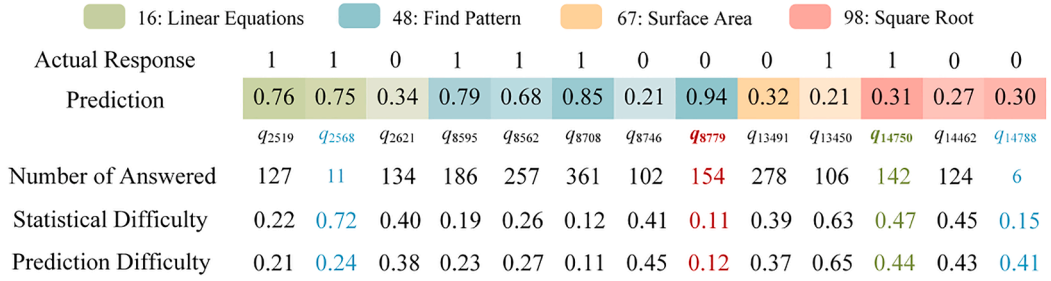


Fig. 9. The learning history and performance predictions of the student with ID 466 on four concepts on ASSIST09. We also provide the number of the question answered by students, statistical difficulty and prediction difficulty of questions.

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets in this paper are public datasets.

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